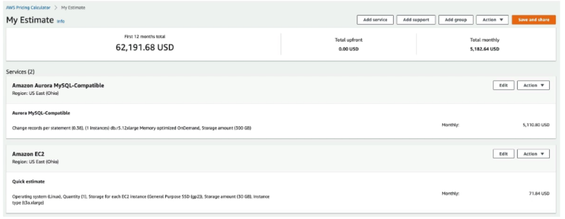
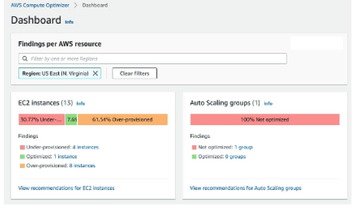
Pricing Calculator: cost estimation for your solution architecture



Saving Plans
- Commit a certain $ amount per hour for 1-3 years
- Compute Saving Plans: EC2, Fargate, Lambda
- Machine Learning

Compute Optimizer
- Reduce costs and improve performance by recommending optimal AWS resources for your workloads
- Helps choose optimal configs
- Uses ML to analyze resources configs and their **utilization CloudWatch metrics**
- Support: EC2, ASG, EBS, Lambda



Estimating Cost

Billing & Cost Management

Tracking Cost

Monitoring against costs plans

Billing Alarms in CloudWatch
- Track overall and per service billing
- For **actual cost**, not for projected costs

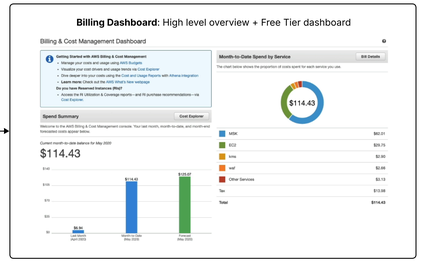
Budgets:
- **Create budget and send alarms when costs exceeds the budget**
- 4 types of budget: Usage, Cost, Reservation, Saving Plans
- For Reserved Instances(RI)
  - Track utilization
  - Supports EC2, ElastiCache, RDS, Redshift

Cost Anomaly Detection
- Continuously monitor using ML
- learns your unique historic spend patterns to detect one-time cost spike and/pr continuous cost increases
- Monitor AWS services, member accounts, cost allocation tags, or cost categories
- Send you the anomaly detection report with root cause analysis
- Get notified using SNS



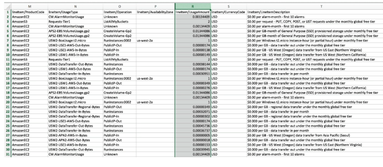
Service Quota
- Notify when you're close to service quota threshold
- Create CloudWatch Alarms on the Service Quotas console
- Request a quota increase or shutdown resources

Billing Dashboard: High level overview + Free Tier dashboard



Cost Allocation Tags: tag resources to create detailed reports
- Tags are used for organizing resources
  - EC2, RDS, VPC
- Resource groups

Cost and Usage Reports - Most comprehensive billing dataset
- Deeper dive into costs and usage
- The most comprehensive set of AWS cost and usage data available, including metadata about AWS services, pricing, reservations
- Cost & Usage Report lists usage for each service category used by an account and its IAM users, and by tags for cost allocation purposes
- Can be integrated with Athena, Redshift, and QuickSight



Cost Explorer: Current and forcast usage
- Visualize, understand, and manage your costs usage over time
- Create custom reports
- Analyze across all acounts
- or monthly, hourly resource level granularity
- Choose Savings Plan
- Forecast usage up to 12 months based on previous usage